

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 01



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Recap of Previous Lecture



Topic

Cyclic group



Topic

Properties w.r.t. cyclic group



Topic

Some important properties



Topics to be Covered



Topic

Representation of graph



Topic

Simple graph, multi-graph and pseudo graph



Topic

Maximum number of edges in a simple graph with n vertices



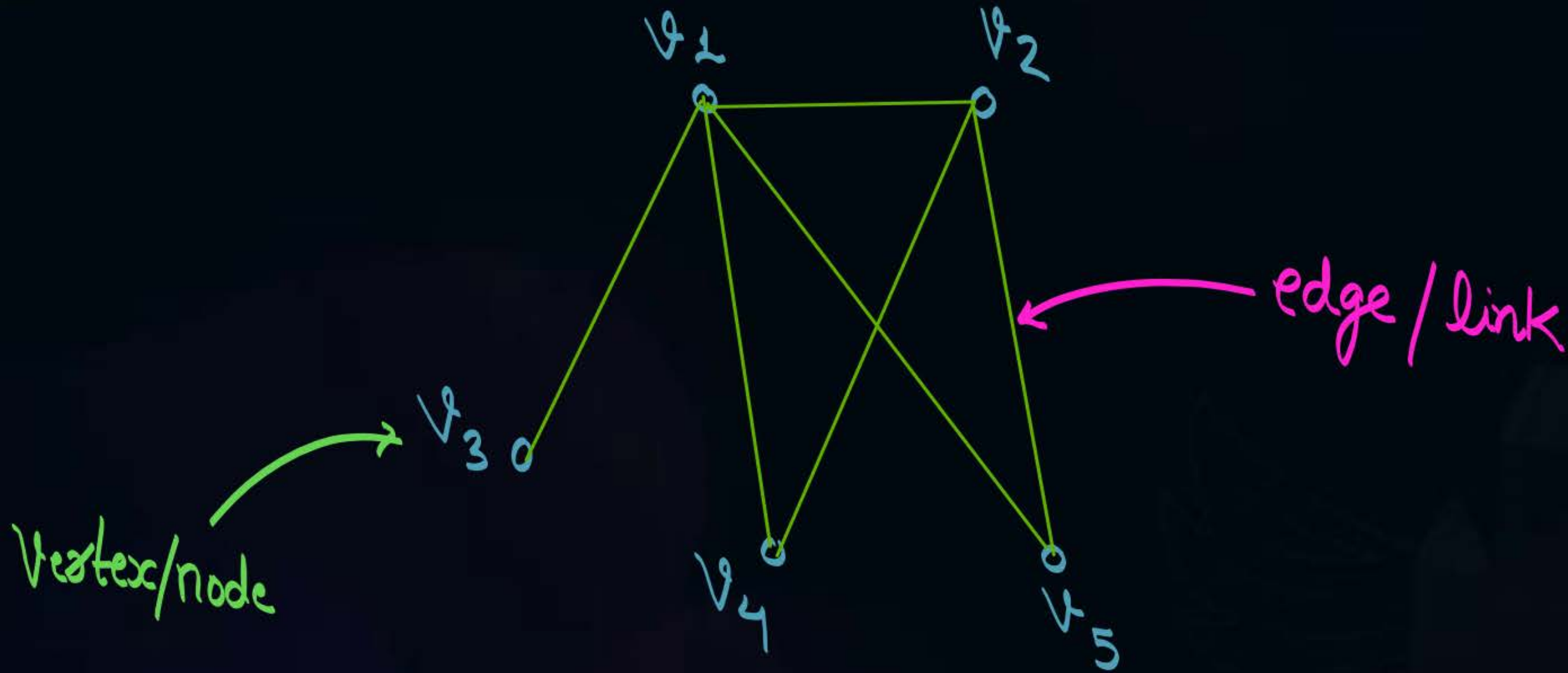
Topic

Number of simple graphs possible with n labeled vertices

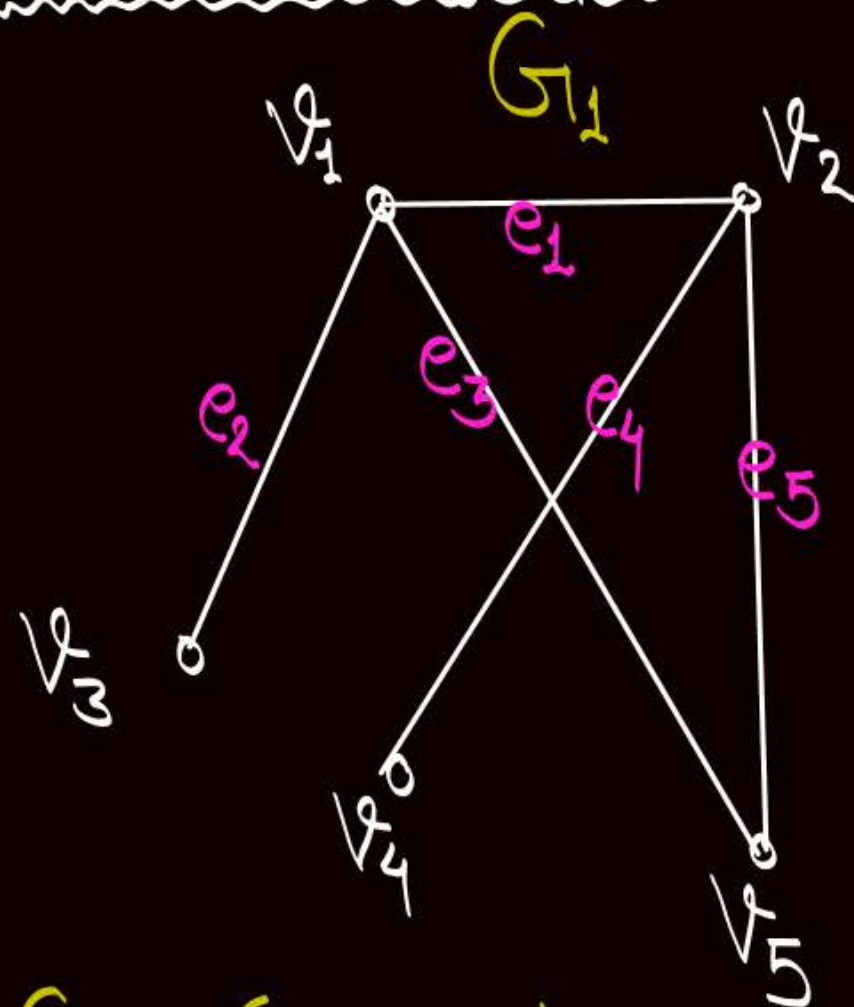




Topic : Graph



* Undirected graph



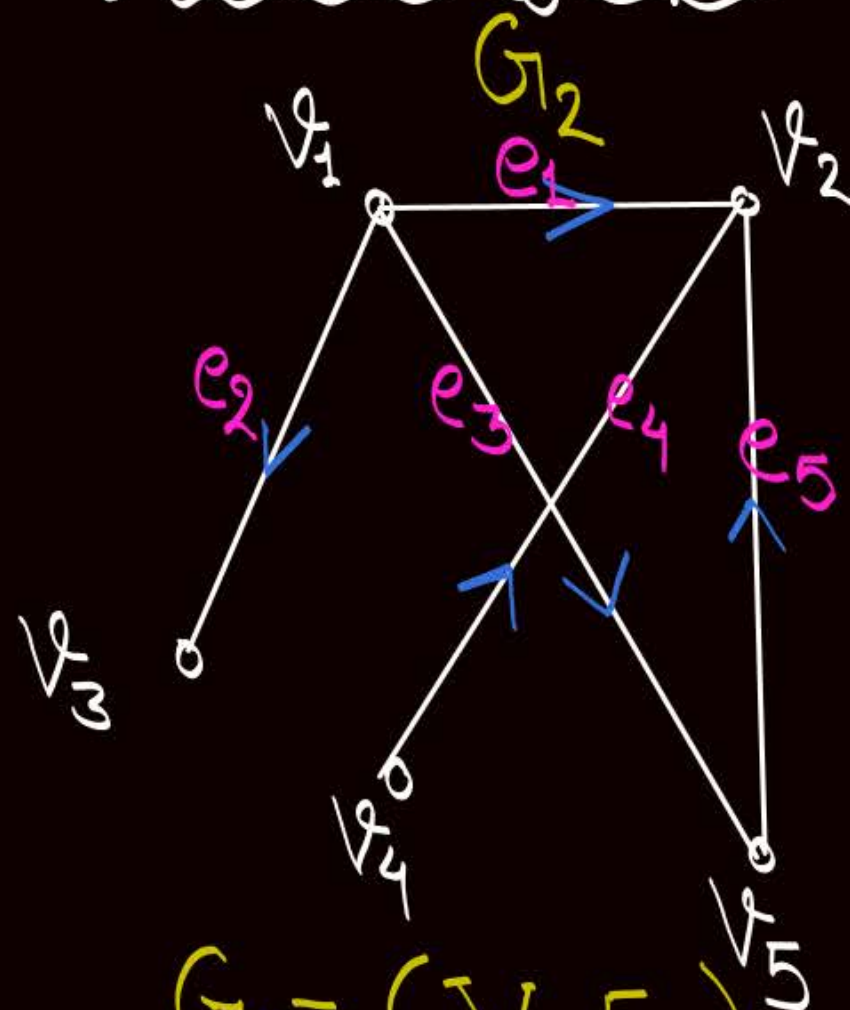
$$G_1 = (V, E)$$

$$V = \{v_1, v_2, v_3, v_4, v_5\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5\}$$

$$= \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_5\}, \{v_2, v_4\}, \{v_2, v_5\}\}$$

Directed graph



$$G_2 = (V, E)$$

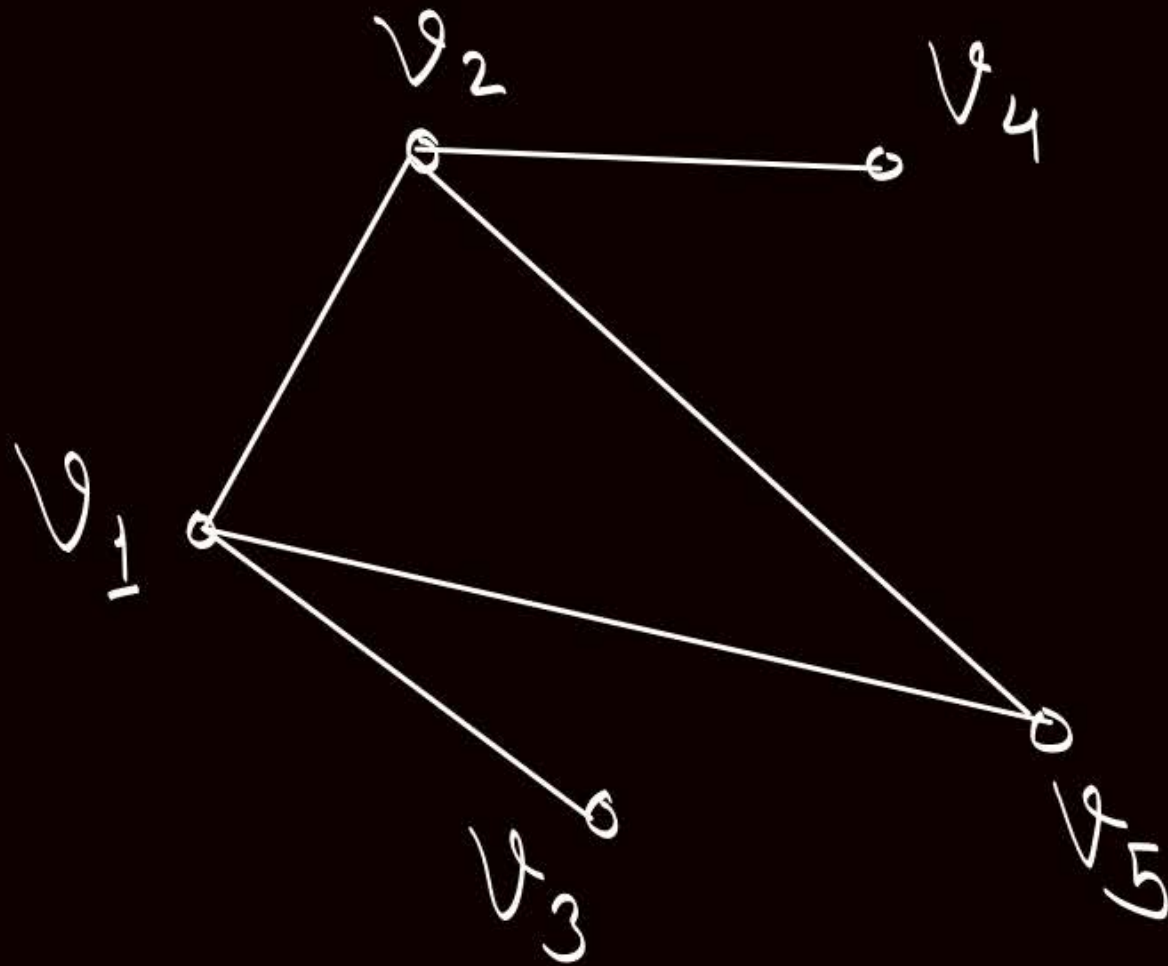
$$V = \{v_1, v_2, v_3, v_4, v_5\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5\}$$

$$= \{(v_1, v_2), (v_1, v_3), (v_1, v_5), (v_4, v_2), (v_5, v_2)\}$$

$$G_1 = (\{v_1, v_2, v_3, v_4, v_5\}, \{e_1, e_2, e_3, e_4, e_5\})$$

$$G_1 = (\{v_1, v_2, v_3, v_4, v_5\}, \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_5\}, \{v_2, v_4\}, \{v_2, v_5\}\})$$





Topic : Graph



i.e. order matter

→ A Graph is defined as an **ordered pair** of two sets,

i.e., $G = (V, E)$

Name
Assigned
to the graph

Set
of
vertices

Set of edges

$|V|$ = No. of vertices (also known as Order of graph G)

$|E|$ = No. of edges (also known as Size of graph G)



Topic : Representation of edge

- ★ In a non-directed graph an edge is represented by a set of two vertices

e.g., $\{v_i, v_j\}$ ^{$= \{v_j, v_i\}$} which represents an edge between vertices v_i and v_j

v_i and v_j are called end vertices associated with an Edge

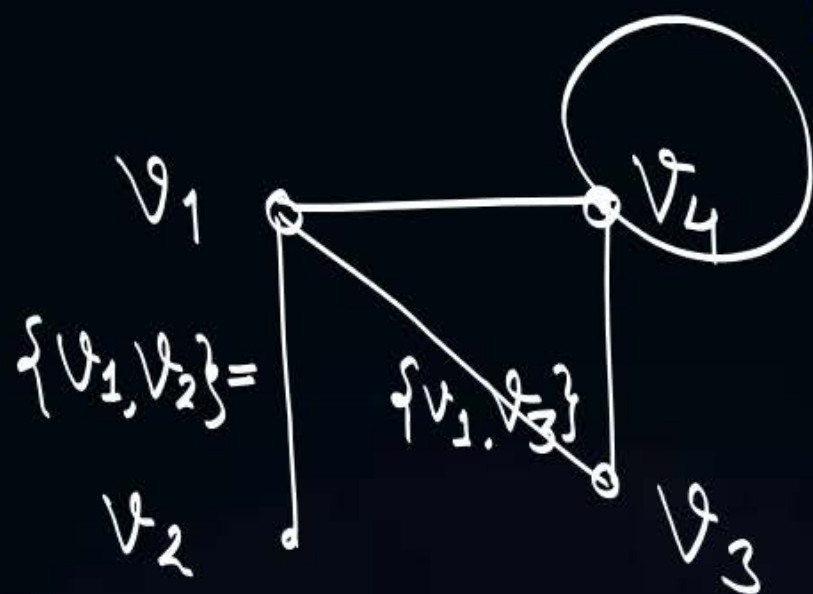
In a directed graph an edge is represented by an order pair of two vertices

e.g., (v_i, v_j) ^{$\neq (v_j, v_i)$} which represents an edge from vertex v_i to vertex v_j

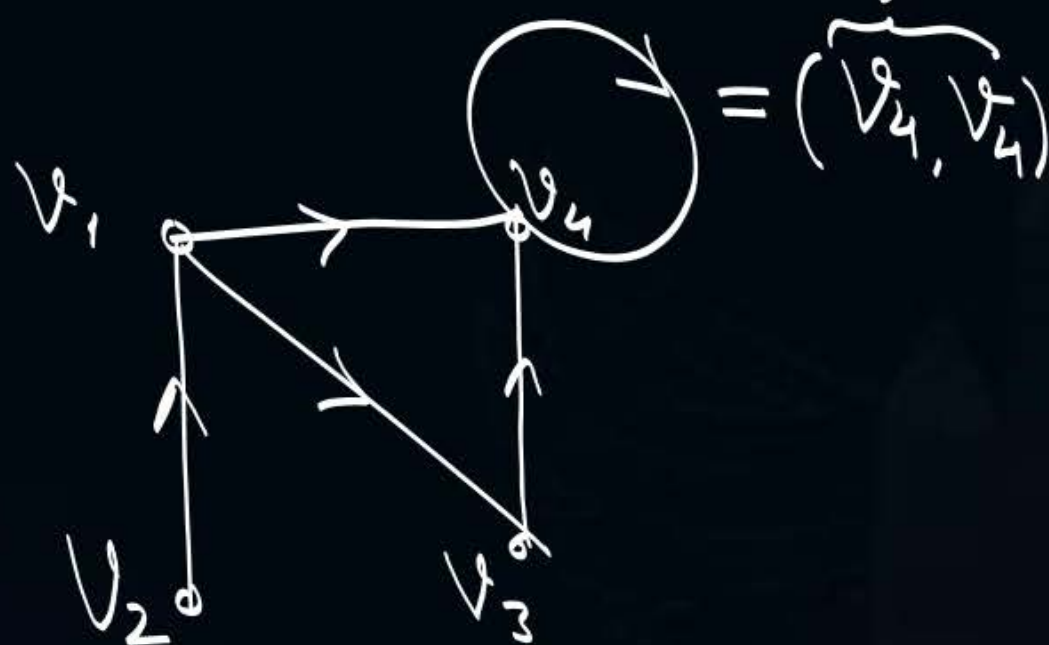


Topic : Loop

Self loop

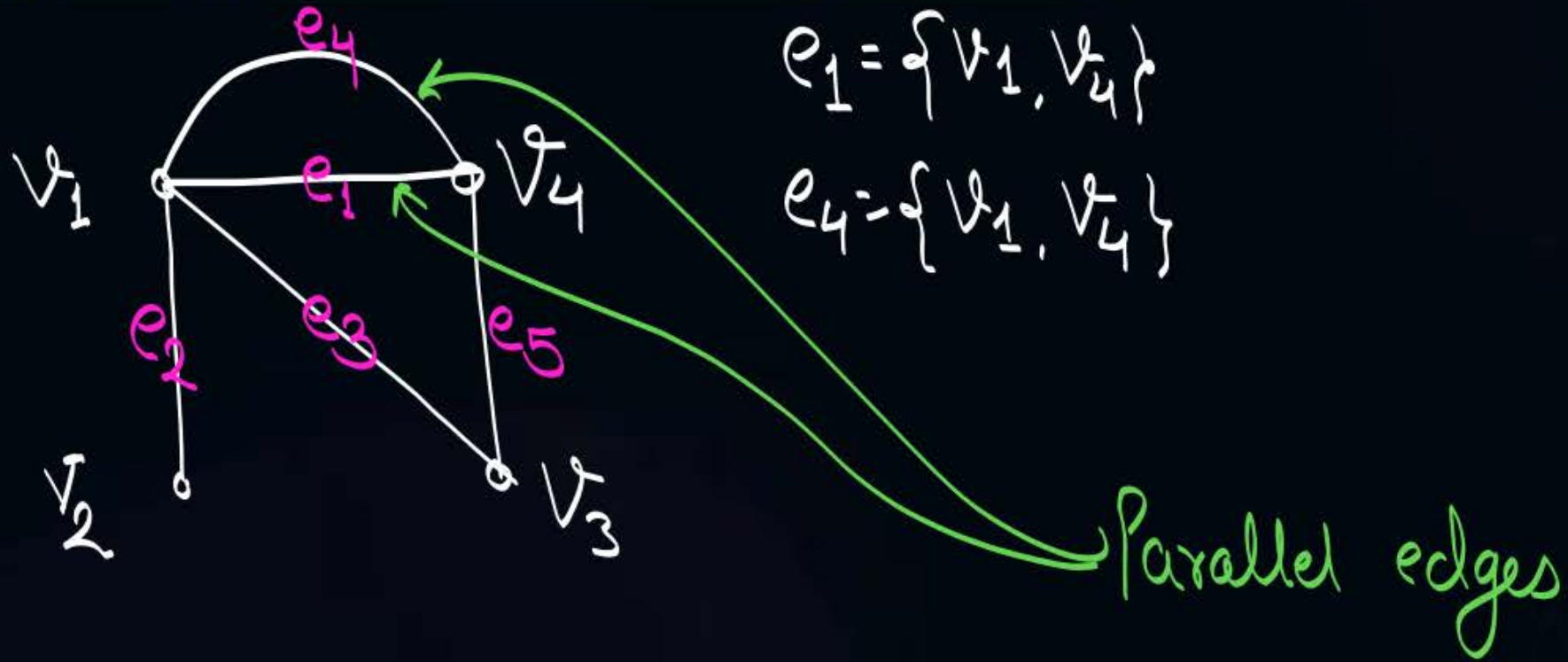


$\{v_4, v_4\}$: Loop is an edge in which both "end vertices" are same





Topic : Parallel Edges





Topic : Parallel Edges

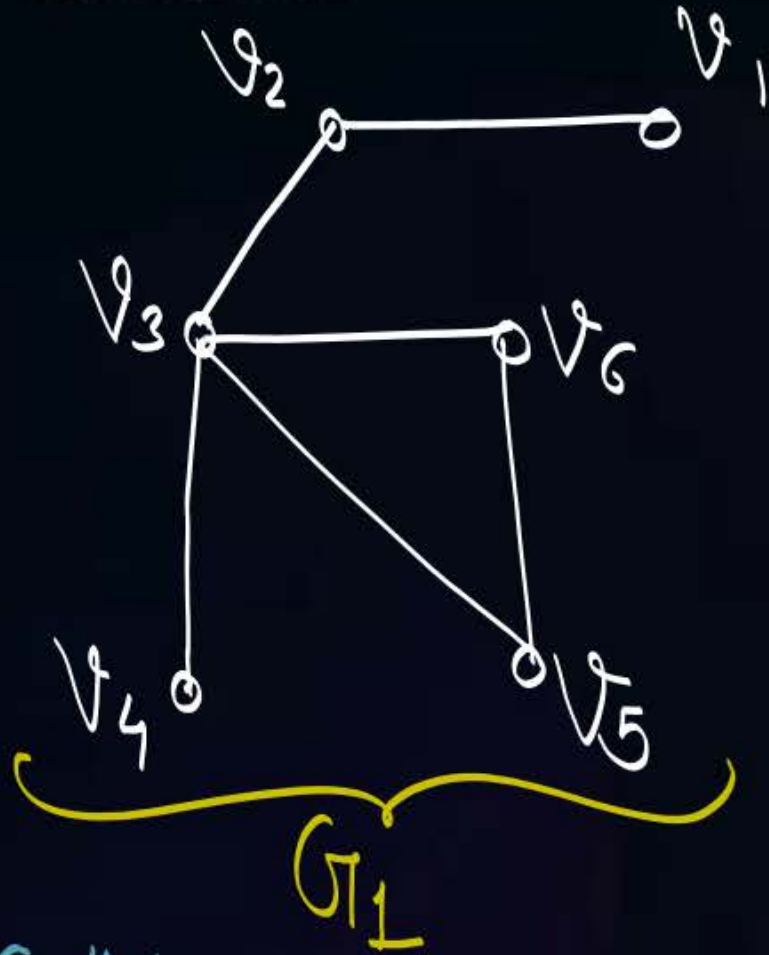


If two or more edge are associated with same end vertices, then edges are called parallel edges and resulting graph is called
multi-graph.

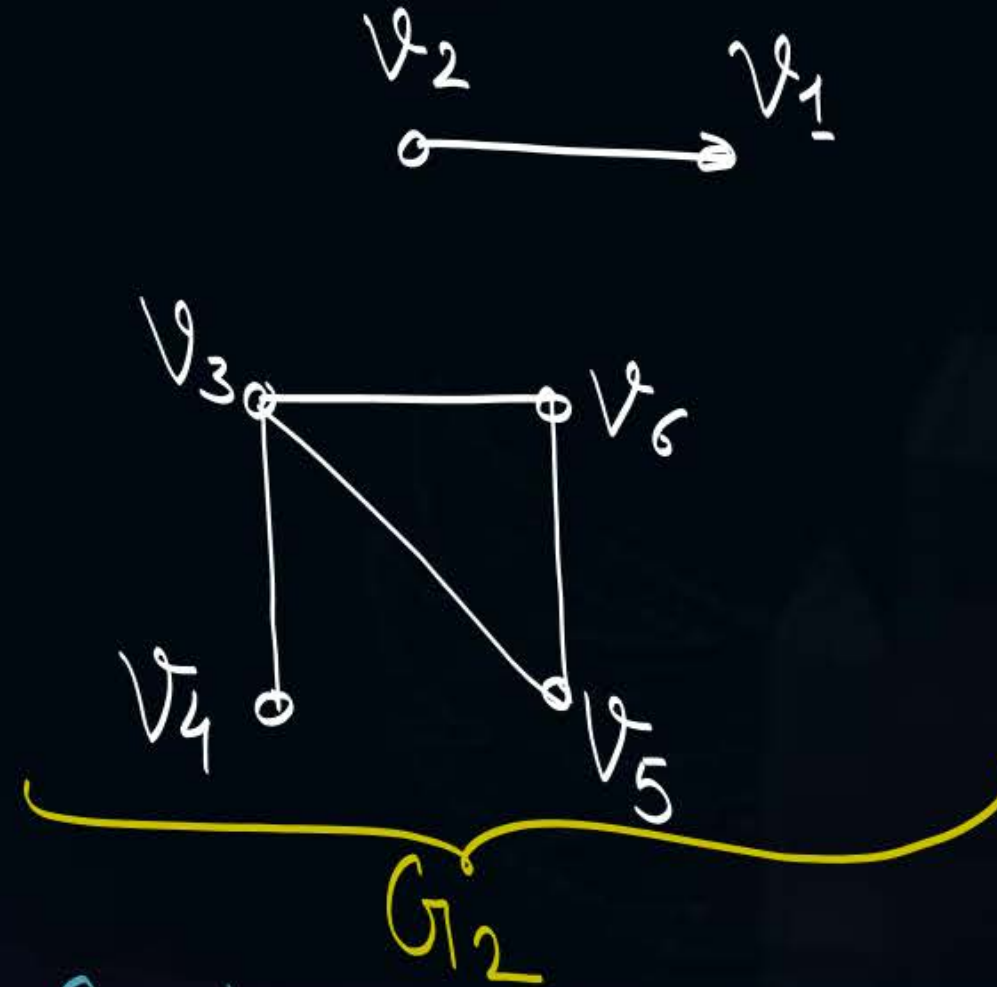


Topic : Simple graph

A graph with no self-loop and no parallel edge is called simple graph.



" G_1 " is a simple graph as well as a connected graph



G_2 is a simple graph but G_2 is a disconnected graph



Topic : Classification of graph

	Loop	Parallel edge
Simple Graph	Not allowed	Not allowed
Multi-graph	Not allowed	Allowed
Pseudo graph	Allowed	Allowed



Topic : Simple Graph

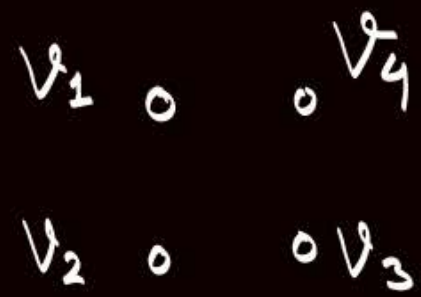
Maximum number of edges possible in "Simple Graph" With n-vertices

- Two vertices are required for an edge. (ie. given two vertices, if we connect them then it is an edge)

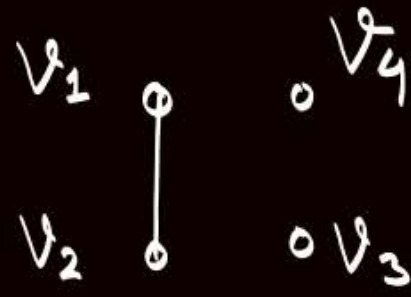
∴ Maximum number of edges possible in a simple graph with n-vertices = Number of ways in which two distinct vertices can be chosen out of n-vertices

$$= {}^nC_2 = \frac{n(n-1)}{2} = \left(\frac{n^2-n}{2}\right)$$

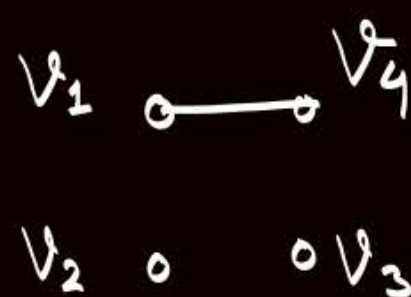
eg: Simple graph with 4-vertices



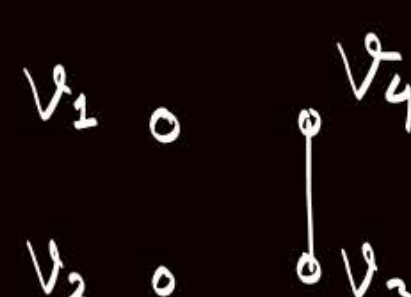
G_1



G_2



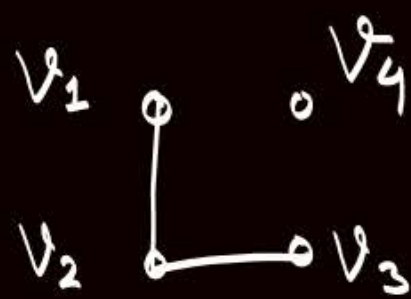
G_3



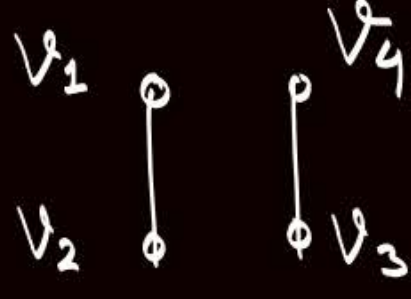
G_4



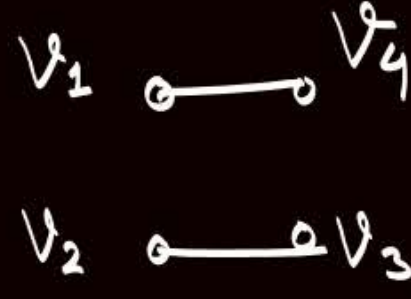
G_5



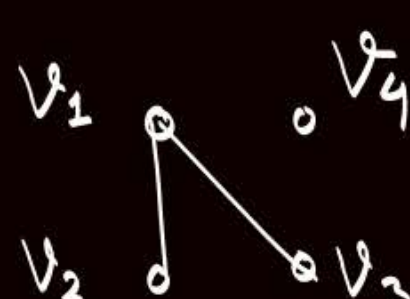
G_6



G_7

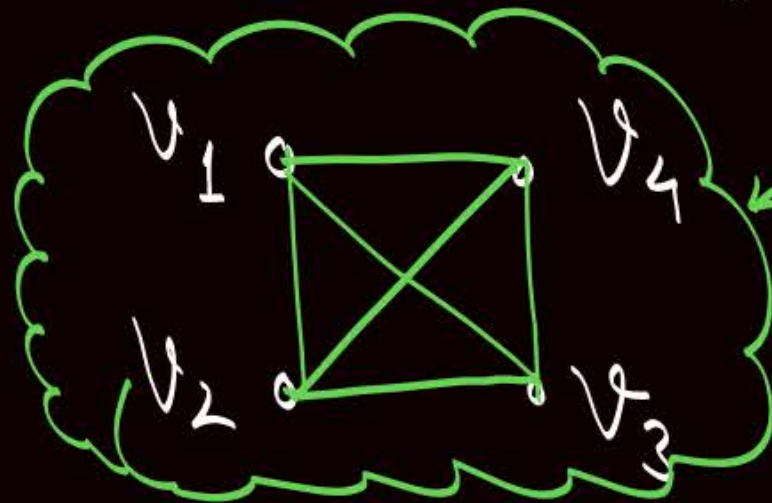


G_8



G_9

... there are many more



With '4' vertices
Maximum $4C_2 = 6$ edges are possible in a simple graph
∴ it is a simple graph with '4' vertices and maximum edges



Topic : Simple Graph

Number of "Simple Graphs" possible With 'n' labeled vertices

- Each simple graph with 'n' vertices will have at most $\left(\frac{n^2-n}{2}\right)$ edges
- $\left\{ \begin{array}{l} \text{Some will have '0' edges} \\ \text{Some will have only '1' edge} \\ \text{Some will have only '2' edges} \\ \text{and so on} \end{array} \right.$

∴ Choose all n vertices

↓
1

And

↓
*

Choose any number of edges out of $\left(\frac{n^2-n}{2}\right)$ edges

$$\left(\frac{n^2-n}{2}\right)C_0 + \left(\frac{n^2-n}{2}\right)C_1 + \left(\frac{n^2-n}{2}\right)C_2 + \dots + \left(\frac{n^2-n}{2}\right)C_{\left(\frac{n^2-n}{2}\right)}$$



Topic : Simple Graph

$$n_{C_0} + n_{C_1} + n_{C_2} + \dots + n_{C_n} = 2^n$$



Number of "Simple Graphs" possible With 'n' labeled vertices

$$\begin{aligned} &= \binom{\frac{n^2-n}{2}}{C_0} + \binom{\frac{n^2-n}{2}}{C_1} + \binom{\frac{n^2-n}{2}}{C_2} + \dots + \binom{\frac{n^2-n}{2}}{C_{\frac{n^2-n}{2}}} \\ &= 2^{\binom{n^2-n}{2}} = 2^{\binom{n}{2}} \end{aligned}$$



Topic : Simple Graph

$$n_{C_0} + n_{C_1} + n_{C_2} + \dots + n_{C_n} = 2^n$$



Number of "Simple Graphs" possible With 'n' labeled vertices

$$= \binom{\frac{n^2-n}{2}}{0} + \binom{\frac{n^2-n}{2}}{1} + \binom{\frac{n^2-n}{2}}{2} + \dots + \binom{\frac{n^2-n}{2}}{K} + \dots + \binom{\frac{n^2-n}{2}}{\frac{n^2-n}{2}}$$

No. of simple graphs possible with n-vertices and at most 'K' edges

No. of simple graphs possible with n-vertices and exactly 'K'-edges

No. of simple graphs possible with 'n' vertices and at least 'K' edges

Q:- How many simple graphs are possible with '5' labeled vertices.

- * In a simple graph with '5' labeled vertices maximum ${}^5C_2 = 10$ edges are possible.
- * Each such simple graph will contain all '5' vertices and at most ${}^5C_2 = 10$ edges

$$\therefore \text{No. of simple graphs possible} = 2^{({}^5C_2)} = 2^{10} = 1024$$

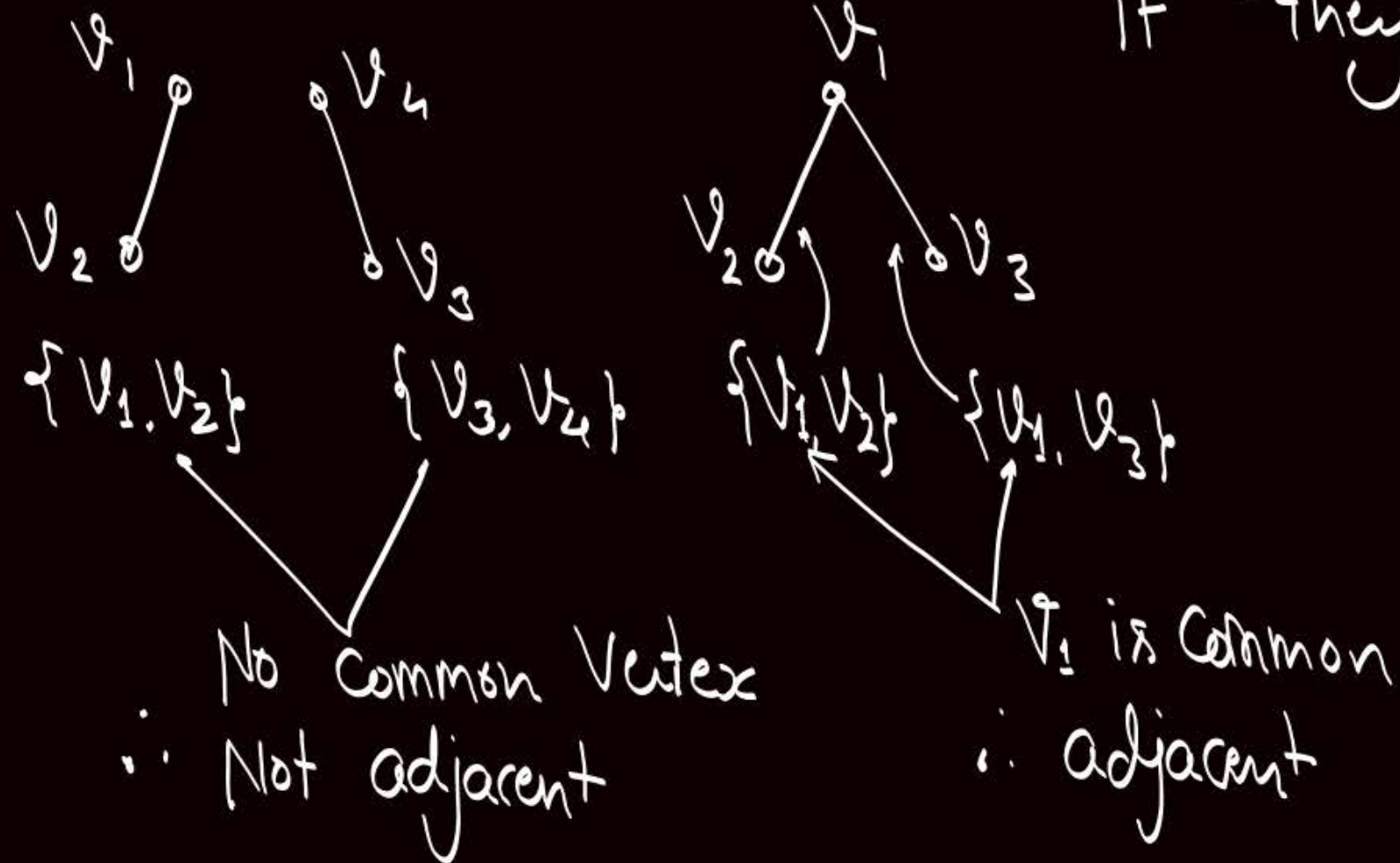
Q: How many simple graphs are possible with '4' labeled vertices and at most '2' edges?

- With '4' vertices, maximum number of edges possible are $= {}^4C_2 = 6$

$$\begin{aligned}\therefore \text{No. of simple graphs possible} &= {}^6C_0 + {}^6C_1 + {}^6C_2 \\ \text{with at most '2' edges} &= 1 + 6 + 15 \\ &= 22\end{aligned}$$

→ Adjacent vertices :— Two vertices are said to be adjacent if there exist a direct edge between them

→ Adjacent edges :— Two edges are said to be adjacent if they share any common vertex





2 mins Summary



✓ Topic

Representation of graph

✓ Topic

Simple graph, multi-graph and pseudo graph

✓ Topic

Maximum number of edges in a simple graph with n vertices

✓ Topic

Number of simple graphs possible with n labeled vertices

THANK - YOU